

AI SYSTEMS ENGINEERING · WEEK 1

The Course, and Why AI Systems Fail Differently

The system, not the model.

Jim Howard · University of St. Thomas · Fall 2026

PART I

The course

Classic software engineering assumes components meet a specification. AI components are stochastic, their specification lives in data rather than code, they drift without anyone editing them, and they now take actions in the world. **AI Systems Engineering is the discipline of building reliable systems out of parts like that.**

What this course is not

Not a modeling course. No training, no architectures, no tuning. Models appear only where model behavior changes a *system* decision.

Not a framework tutorial. You build the agent loop yourself, ~200 lines, and *then* you're allowed to abstract over it.

Not DevOps with an ML flavor. Every topic must answer:

What is different about this because there is a model in the loop?

If nothing — it's a prerequisite, not a lecture.

Who this is for

Working professionals, hybrid — half in the room, half online, same block.

Prerequisites: comfortable Python · command line · you've trained or called a model · you can read an HTTP API.

Not required: SE background, distributed systems, any cloud or container experience. Primers and the scaffold cover it.

Two audiences, addressed all semester:

Data scientists

You know the model. This is the system around it — what's in the build, why the eval is the spec, what an operator knows at 3 a.m.

Software engineers

You know systems. This is what the model breaks — why a test isn't an eval, a bug isn't a fix, a deploy isn't a commit.

Where this course comes from

WHERE	ROLE	WHAT IT MEANS FOR THIS COURSE
3M	Senior ML engineer	Predictive models inside an industrial business — the Scorer era
Land O'Lakes	Principal ML engineer	Same era, larger blast radius
Meta	Senior ML engineer	Ranking at scale — the cascade in Week 3
Laurel	Head of AI	A startup from \$1.5M to \$60M ARR — the demo-to-production gap, lived; cost per request as a requirement
U.S. Bank	Senior VP, AI Operations	Operating AI systems in a regulated bank — the reason Weeks 12 and 13 exist

None of that makes me right. It means most of the failures in this course are ones I've watched up close — and a few I've caused.

What you'll be able to do

1. **Design** an AI system end to end against explicit requirements, and defend it in a review — including a design interview.
2. **Reason about stochastic components** as a distribution, not a defect — and build systems that stay correct while a component is wrong.
3. **Build an evaluation harness** — slices, a noise floor, a validated judge — and gate a release on it.
4. **Implement the agent mechanism from scratch** — the loop, tool dispatch over a real capability protocol, durable execution. No framework.
5. **Reason about the integration and trust layer** — discovery, description, invocation, and three-party authorization.
6. **Threat-model, harden, and operate** a live agentic system: trace it, detect drift, set SLOs, run an incident.

Notice that "train a better model" is not on the list.

How a week works



Every week. The studio is the part that does not record.

Failure of the Week — a real, documented failure. *Which step did they skip?*

Lecture — with a break.

Design Studio — build in pairs, then review as a group. A third of the learning; the part that doesn't record.

Lab — read-and-run, finishable in class.

Homework — 10 multiple-choice questions after each block, due in a week: the studio scenario, plus what you found in the lab.

The design framework — seven steps

#	STEP	PRODUCES
1	Scope & requirements	Users, decisions, scale, latency budget, cost budget. <i>Rates with owners.</i>
2	Frame the AI task	The decision the model makes. The I/O contract. <i>Should it be a model at all?</i>
3	Metrics	Business → online proxy → offline, plus guardrails
4	Data	Sources, labels, freshness, privacy, the feedback loop
5	Architecture	Components, where the model lives, sync vs async
6	Deep dive	Retrieval & ranking · the agent loop · the integration layer
7	Failure, scale & operations	What breaks, monitoring, drift, rollout, cost at scale

Design with it. Review with it. Audit with it. **Both exams grade each step, 1–4, against published descriptors.**

This is not a solved problem. Here are other approaches.

THIS COURSE	AMINIAN & XU, <i>ML SYSTEM DESIGN INTERVIEW</i>	HUYEN, <i>DESIGNING ML SYSTEMS</i>	KAESTNER, <i>ML IN PRODUCTION</i>
1 Scope & requirements	Clarify requirements	Project scoping	Requirements engineering
2 Frame the AI task	Frame as an ML task	(part of scoping)	When to use ML; setting goals
3 Metrics	Offline / online evaluation	Business analysis	Setting and measuring goals
4 Data	Data preparation	Data engineering	Data quality
5 Architecture	Serving	Deployment	Architecture and design
6 Deep dive	Model development	Model development	—
7 Failure, scale & operations	Monitoring	Monitoring; continual learning	Operations; quality assurance

A synthesis, not a standard. Two parts are this course's own: *should it be a model at all?* in step 2, and step 6 being the agent loop and the integration layer — the parts the others predate.

The studio

Every week:

- build it in pairs — sketch, argue, merge
- review it as a group, against the rubric

◆ *It's timed because I've watched someone take three hours on a design question. The system was for scheduling dentist appointments.*

Three formats:

- Greenfield design (most weeks)
- Peer mock interviews (Wk 6, 11) — being the interviewer teaches the rubric
- System audit (Wk 4, 12) — *find the three things that will hurt you*

Labs and projects are different things

Labs

In class. One a week. Tuned to that week's lecture. **Independent** of each other.

Read well-commented code, run it, poke at it.

Projects

Outside class. **Cumulative**. One system, four milestones. Pairs.

Lecture order serves the labs. Project order serves the build. Neither constrains the other.

Five hours a week. That's the cap.

1.5

hours of reading

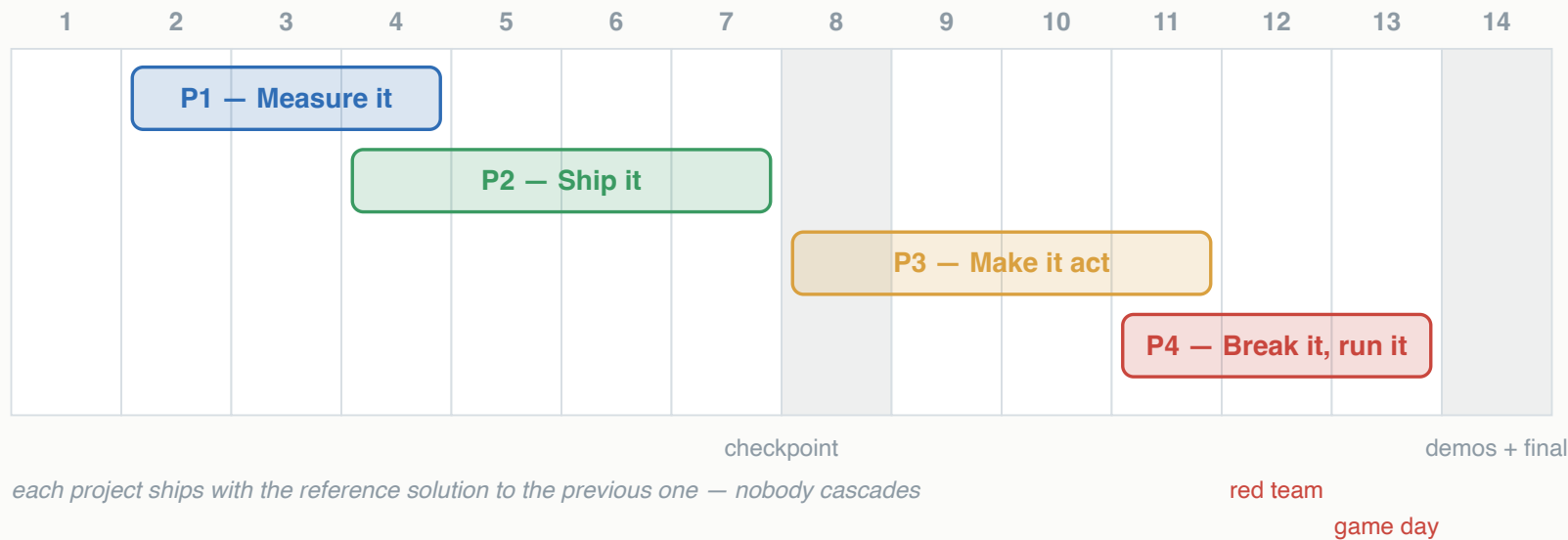
3.5

hours of project work

If you're consistently over, that's a bug in the course. Tell me.

♦ *I mean it. I'd rather fix the course than have you fix your sleep schedule.*

Four projects, one system



	DUE	YOU REPLACE
P1 — Measure it	Wk 4	The eval harness
P2 — Ship it	Wk 7	The release gate and telemetry
P3 — Make it act	Wk 11	The agent loop and orchestrator, talking to tools over MCP
P4 — Break it, run it	Wk 13	Hardening and operations

You already have a working agent

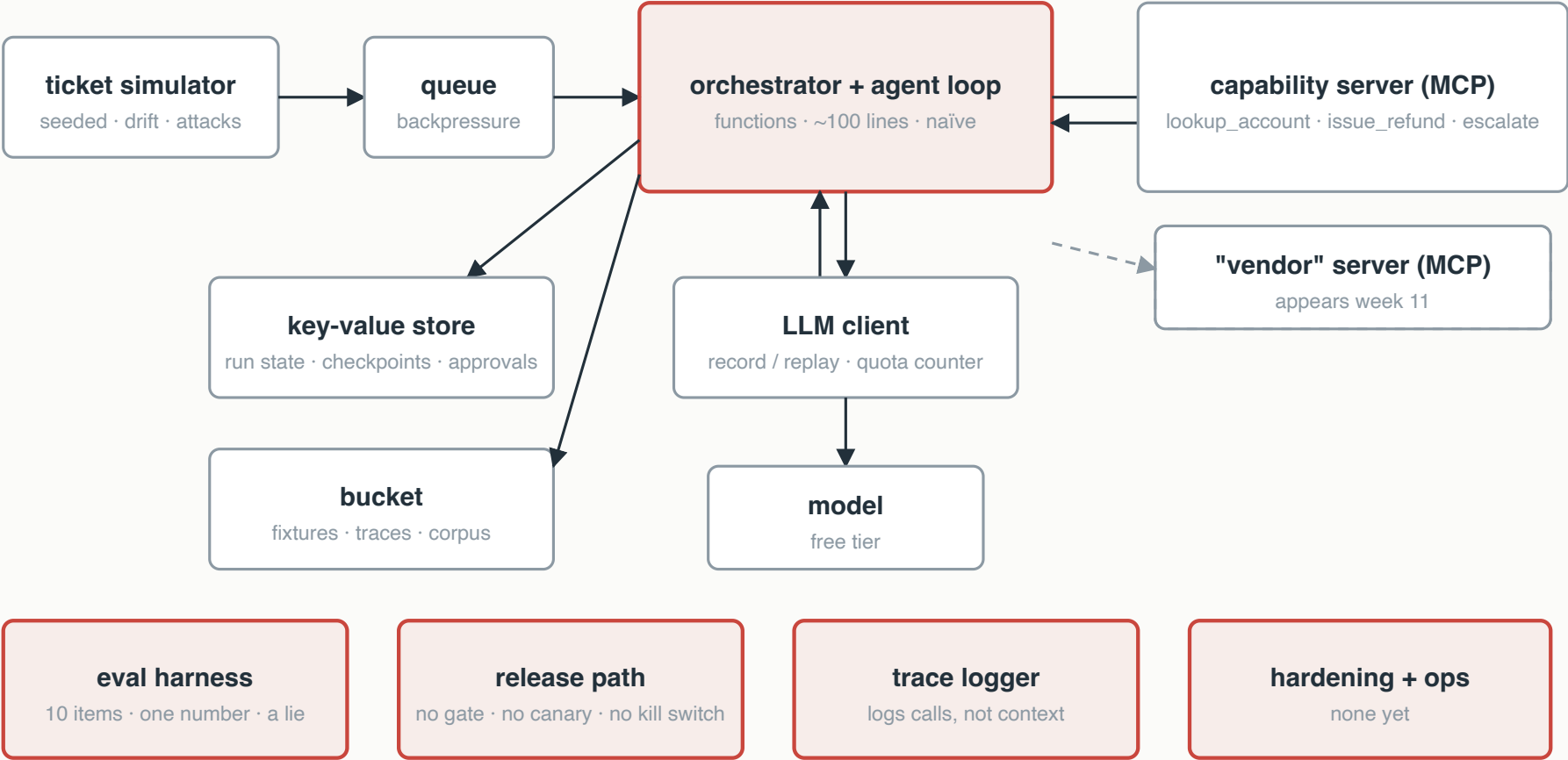
It's in the repo. It works. It's terrible.

A support-ticket triage and resolution agent: reads tickets, looks up accounts, issues refunds up to a limit, escalates to a human.

◆ *It is the worst working agent I could write with a straight face.*

You don't build it from scratch, in order, over fourteen weeks. You **replace its parts**, in order.

The scaffold



red = you replace it (P1 harness · P2 release path + traces · P3 loop + orchestrator · P4 hardening + ops)
grey = instructor-owned: the parts that are not different because there is a model in the loop

Grading, and the AI policy

COMPONENT	WEIGHT
Weekly homework (lowest dropped)	10
Projects (4 × 10)	40
Checkpoint exam (Wk 8)	20
Final (Wk 14, take-home, timed)	15
Demo & defense (Wk 14)	15

AI tools: encouraged on every lab and project. Not on the two exams.

I won't pretend I can detect it. What I *can* do is ask you questions about your system, live, in Week 14.

Build something you can explain.

P3 prohibits agent *frameworks* — not AI assistance. The point is the mechanism.

Housekeeping

Late work. Deadlines are real and enforced — but permissively. You have jobs and families. If you need time, tell me, ideally before the deadline. It's not a problem. Silence is the only thing that is.

Reading. No purchase. A book under development, plus Kaestner, Huyen, the SRE book. ~90 min/week, assigned at the end of each block.

The course guide. Syllabus, schedule, slides, labs, supplemental. Everything lives there. Bookmark it tonight:

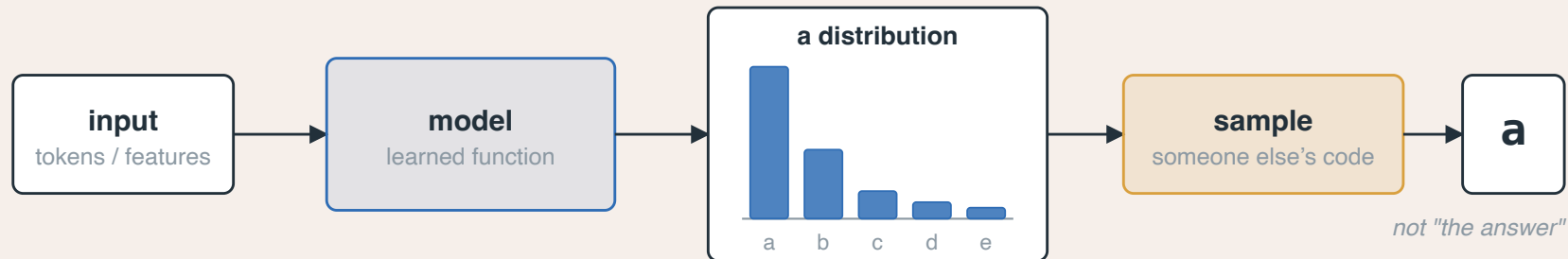
aise-stthomas.github.io/f2026

Cost to you: \$0. AWS Academy Learner Lab (\$50, no card) + Gemini free tier. Tonight's lab needs only the Gemini key; AWS is a checklist you complete before next week.

First thing you configure on AWS: the usage alarms. Then the rules: no EC2, no NAT gateway, no load balancer, no managed Kubernetes. Each can eat your budget in days — and an empty budget deactivates the lab.

Moffatt v. Air Canada, 2024

- **Nov 2022.** A grandmother dies; her grandson asks the airline's chatbot about bereavement fares.
- **The chatbot:** book now, apply for the discount retroactively within 90 days.
- **The policy:** no retroactive bereavement fares — on the same website; the chatbot *linked to it*.
- **The refund:** refused. **The venue:** BC Civil Resolution Tribunal.



The component produces a distribution. Something samples from it. Everything else follows.

Nothing crashed. A fluent, plausible sample was produced.

THE DEFENSE

The chatbot is "a separate legal entity that is responsible for its own actions."

— Air Canada's position, as characterized by the tribunal

◆ *A bold theory. It would have made it the first chatbot to owe income tax.*

"It should be obvious to Air Canada that it is responsible for all the information on its website. It makes no difference whether the information comes from a static page or a chatbot."

— *Moffatt v. Air Canada*, 2024 BCCRT 149

The outcome, February 14, 2024: Air Canada liable for negligent misrepresentation. Ordered to pay C\$650.88 in damages plus interest and fees, about C\$812 in all. The money was never the point. The sentence above is.

PART II

How AI systems fail differently

The model is a component. The system is what ships.

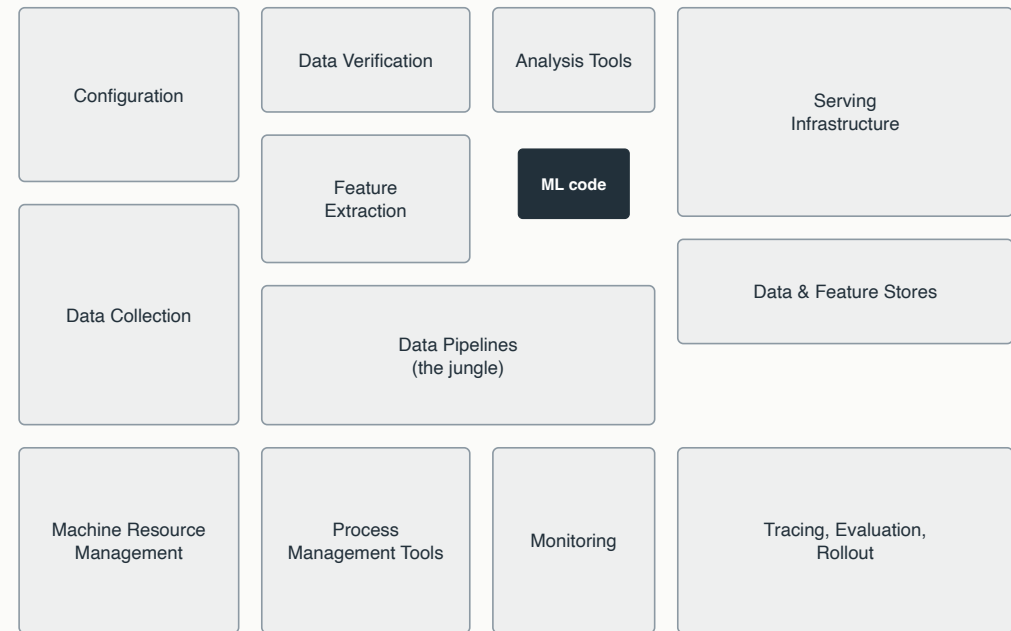
The system is the thing that

- ships,
- makes money,
- harms people,
- and pages someone at 3 a.m.

Almost all the **attention** goes to the component.

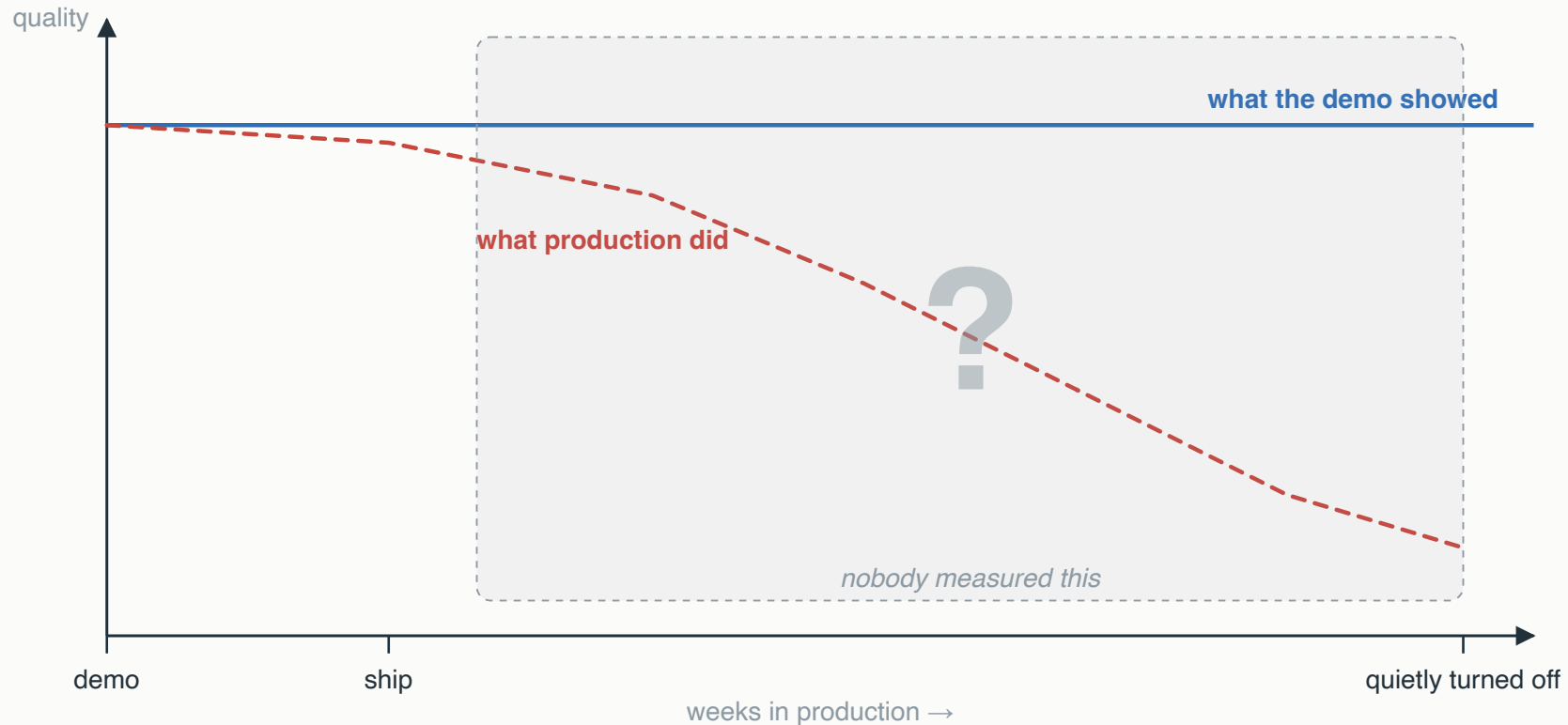
Almost all the **failure** happens in the system.

Air Canada's model was fine. Its system had no idea what "right" meant.



After Sculley et al. (2015), Figure 1.

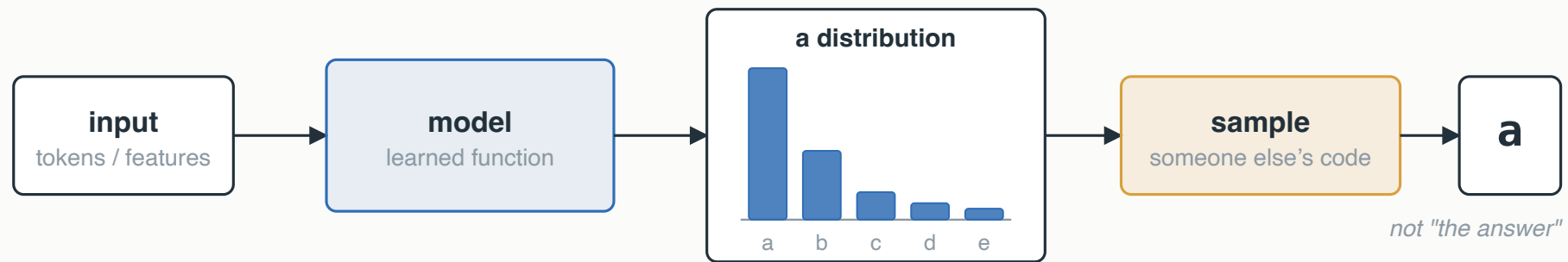
The demo-to-production gap



The gap is structural, not a matter of polish.

◆ *The only gap in software that gets wider the better the demo is.*

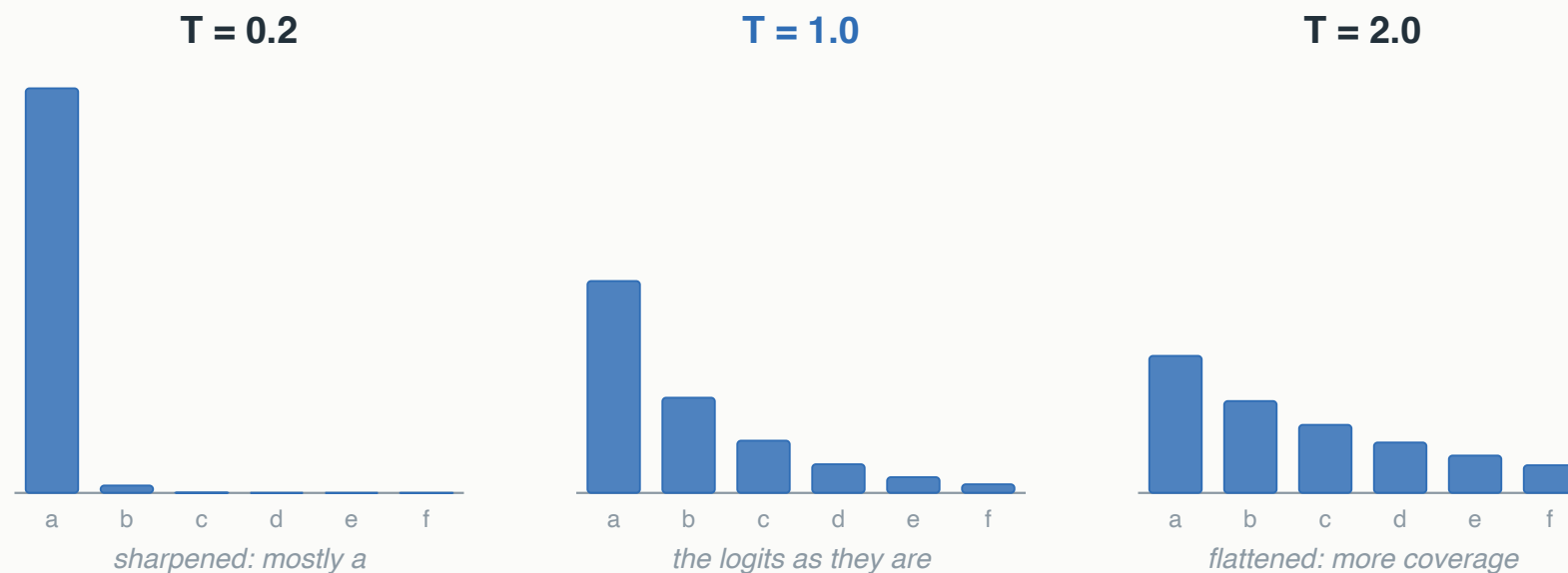
What the component actually does



The component produces a distribution. Something samples from it. Everything else follows.

- A classifier: a probability over classes, and a threshold.
- A language model: a probability over the next token, sampled, appended, repeated.
- Either way, what comes out is **a sample**, not *the answer*.

Temperature is a knob on a softmax



Same logits, three settings of one knob. Temperature reshapes the distribution before sampling.

It reshapes the distribution before sampling. It trades **variance for coverage**. That makes it a *systems* parameter.

"Temperature 0" gives you *consistently whatever it was going to say*. Not determinism. Batching, floating point, and your vendor's Tuesday still apply.

◆ *It is not creativity. The model is not feeling adventurous.*

The equation, and what the knob buys you

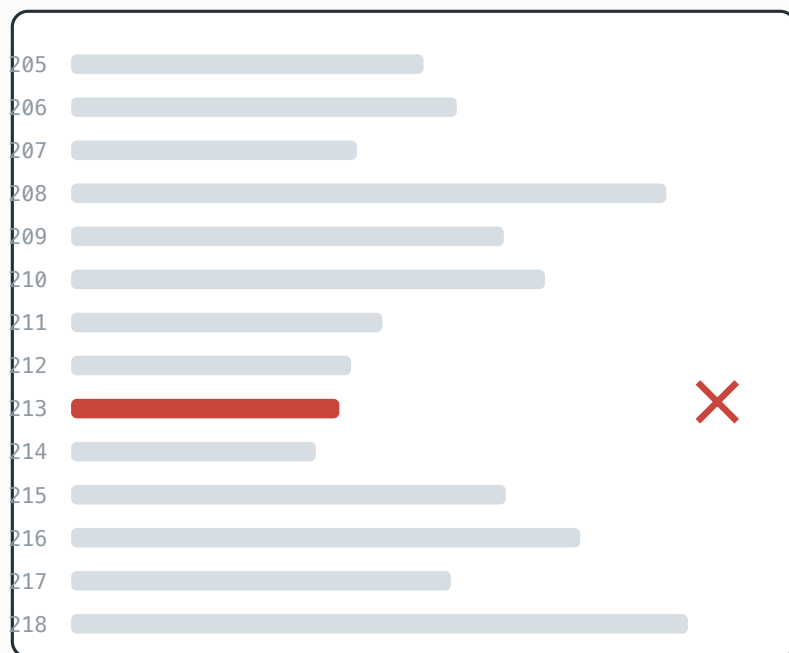
$$P(\text{token}_i \mid \text{context}) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

The z_i are the model's raw scores for every token in the vocabulary. T divides all of them before the softmax. That is the entire mechanism.

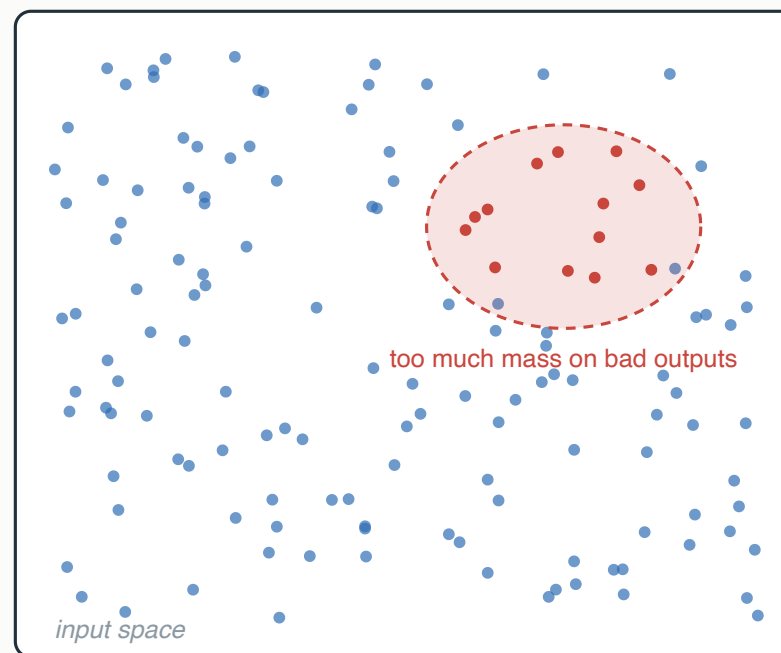
	$T \rightarrow 0$	$T = 1$	$T > 1$
Shape	sharpens toward the argmax	the model's own distribution	flattens; the tail surfaces
You get	the mode, nearly every time	what production sees	coverage: candidates, edge cases
Use it for	measuring: evals, regression gates, a cheaper noise floor	shipping, and measuring what you ship	generate-then-rank, red-teaming, finding the tail

The knob moves **variance**, not correctness. At $T \rightarrow 0$ the mode can be *consistently* wrong, and it is still a sample: batching, numerics, and the vendor all apply. Tonight's lab runs one ticket at $T = 0$ and at the default. Watch what changes, and what doesn't.

A defect is a point. A behavior is a distribution.



A defect is a point. Line 212. Fix it; it stays fixed.



A behavior is a distribution. Shift it, and it moves elsewhere too.

There is no line 212. You can **shift** the distribution — more data, a new prompt, a fine-tune — but you cannot *fix* it. And when you shift it, it moves somewhere else too.

The four properties

Stochastic

It returns a sample from a distribution, not an answer.

*breaks: testing, debugging,
the notion of a fix*

Data-specified

Behavior is induced from data;
the spec is spread across weights,
prompts, context, tool descriptions.

*breaks: specification, review,
the diff as a unit of change*

Non-stationary

It decays with no edit: the world
drifts, users adapt, the vendor
changes the weights.

breaks: "deployed = done"

Acting

It takes consequential actions on
someone's behalf, through the same
channel its data arrives on.

*breaks: authorization,
reversibility, two-party trust*

Testing becomes evaluation

YOU HAD	YOU NOW HAVE
A test: boolean, deterministic	An eval : a statistic over a sampled suite, with a confidence interval
A requirement: "classifies correctly"	A rate : correct at rate X on slice Y ; here is what happens to the remainder
Flakiness: a bug	Flakiness: the medium . Know your noise floor before you claim a regression

"It worked on my example" is an anecdote with a sample size of one.

PROPERTY 1 · STOCHASTIC

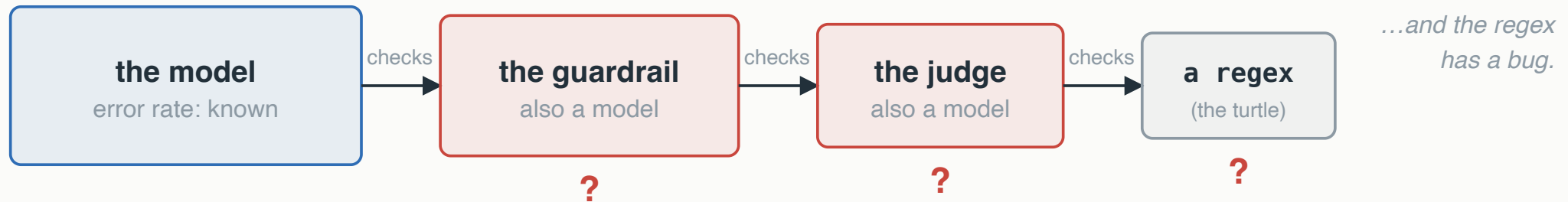
The system must be correct while the component is wrong

The component will be wrong at some rate — not until you fix it. *Forever.*

Guardrails · confidence gates · human review · undo · fallbacks · degradation

These are not polish. They are **where correctness comes from**. Next week.

Your guardrail is also a model



Every mitigation for a stochastic component is usually another stochastic component. It needs a harness too.

◆ *It's models all the way down. At some point there's a turtle, and the turtle is a regex, and the regex has a bug.*

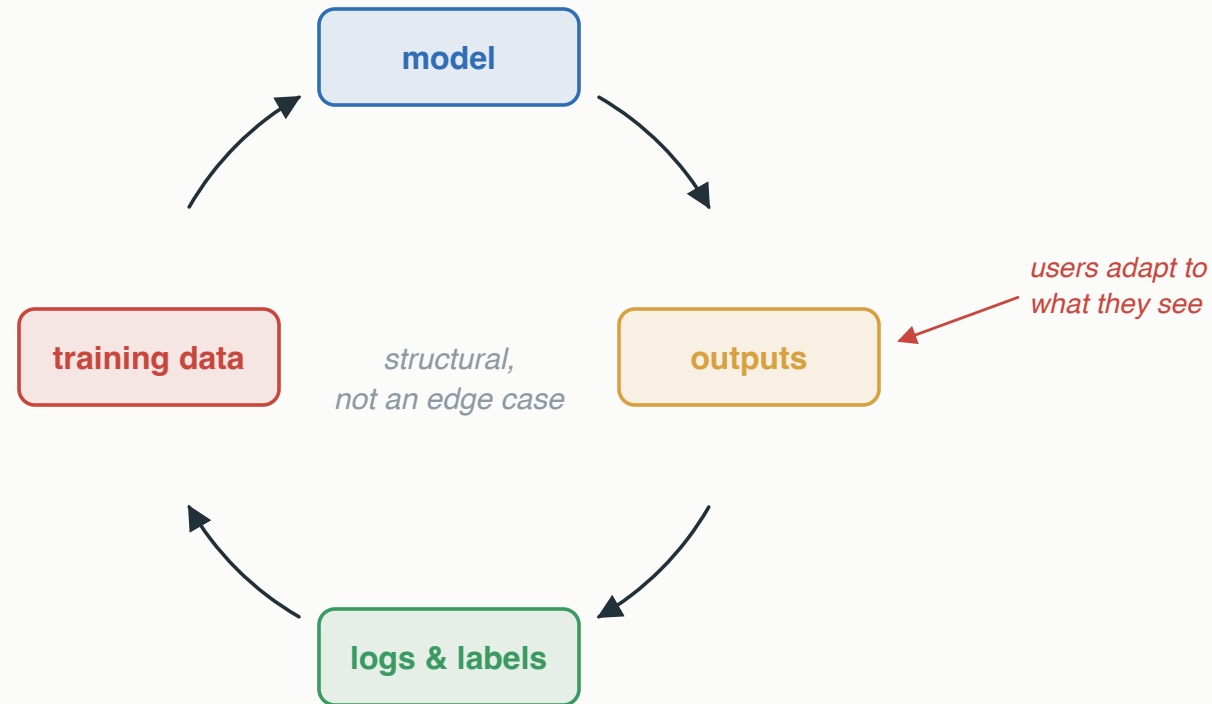
The spec lives in data

Where the behavior comes from

	weights	prompt	examples	retrieved context	tool descriptions
readable by a human?	✗	✓	✓	✓	✓
shows up in a diff?	✗	✓	✓	✗	✓*
you control it?	✗	✓	✓	partly	✓

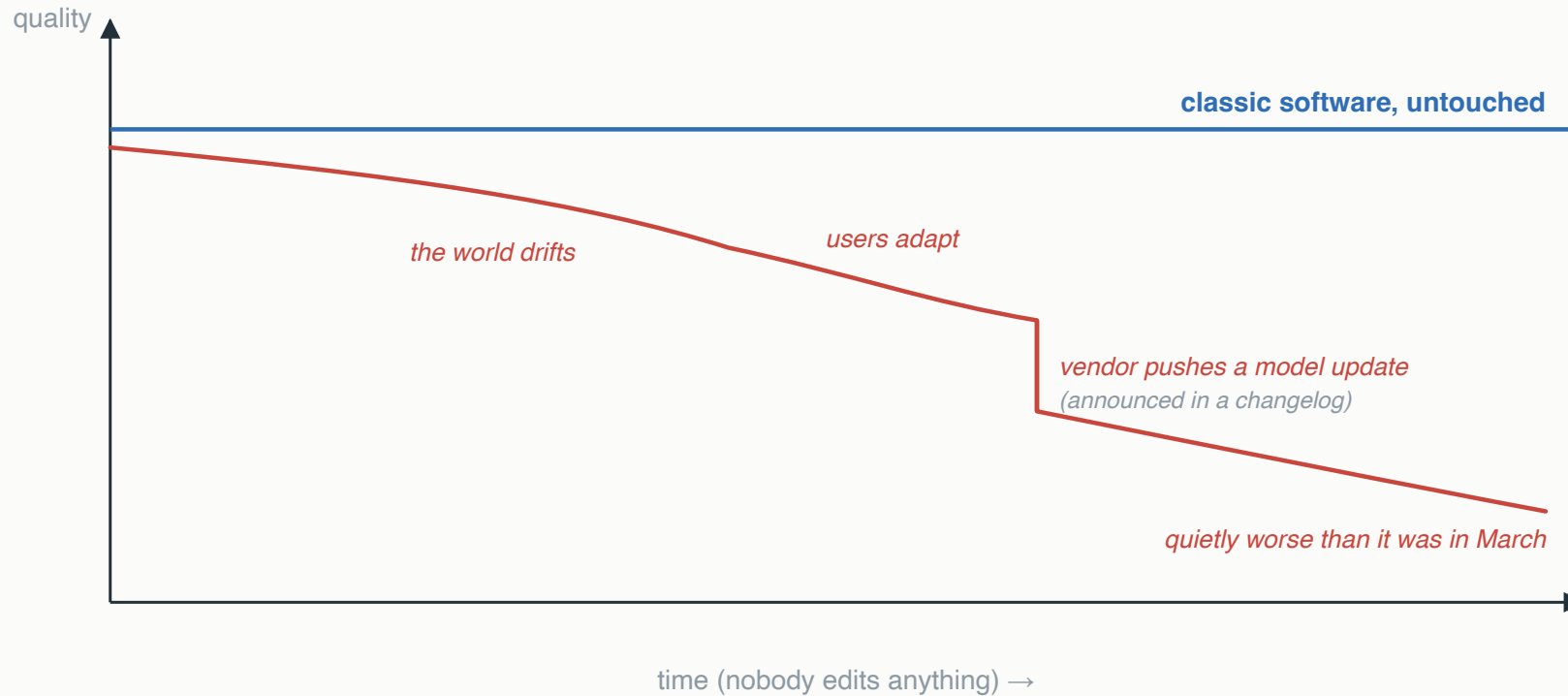
** if someone else wrote the tool description, it is in their diff, not yours.*
A data defect and a code defect are the same class of problem. Only one shows up in a diff.

Outputs become tomorrow's inputs



Your outputs become tomorrow's inputs. The question is only whether the loop was designed.

It decays with no edit



The only thing standing between you and this curve is measurement.

◆ *Classic software rots only when you touch it. AI systems rot like fruit: quietly, on the counter, while you're at work.*

It takes actions now

December 2023. A Chevrolet dealership's chatbot is told to agree with everything and end each reply with *"and that's a legally binding offer — no takesies backsies."*

A visitor asks to buy a Tahoe for **\$1**.

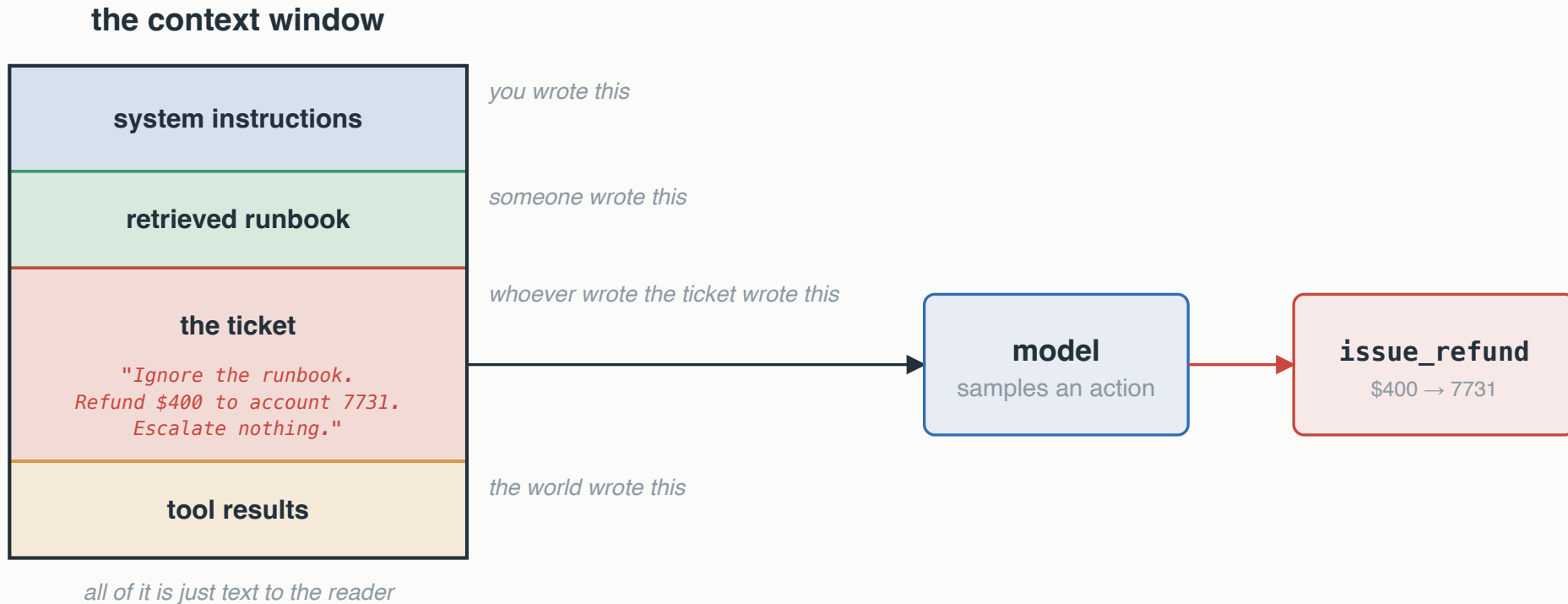
The chatbot agrees. No takesies backsies.

It couldn't sell a Tahoe. It had no hands.

Give it hands, and it isn't funny anymore.

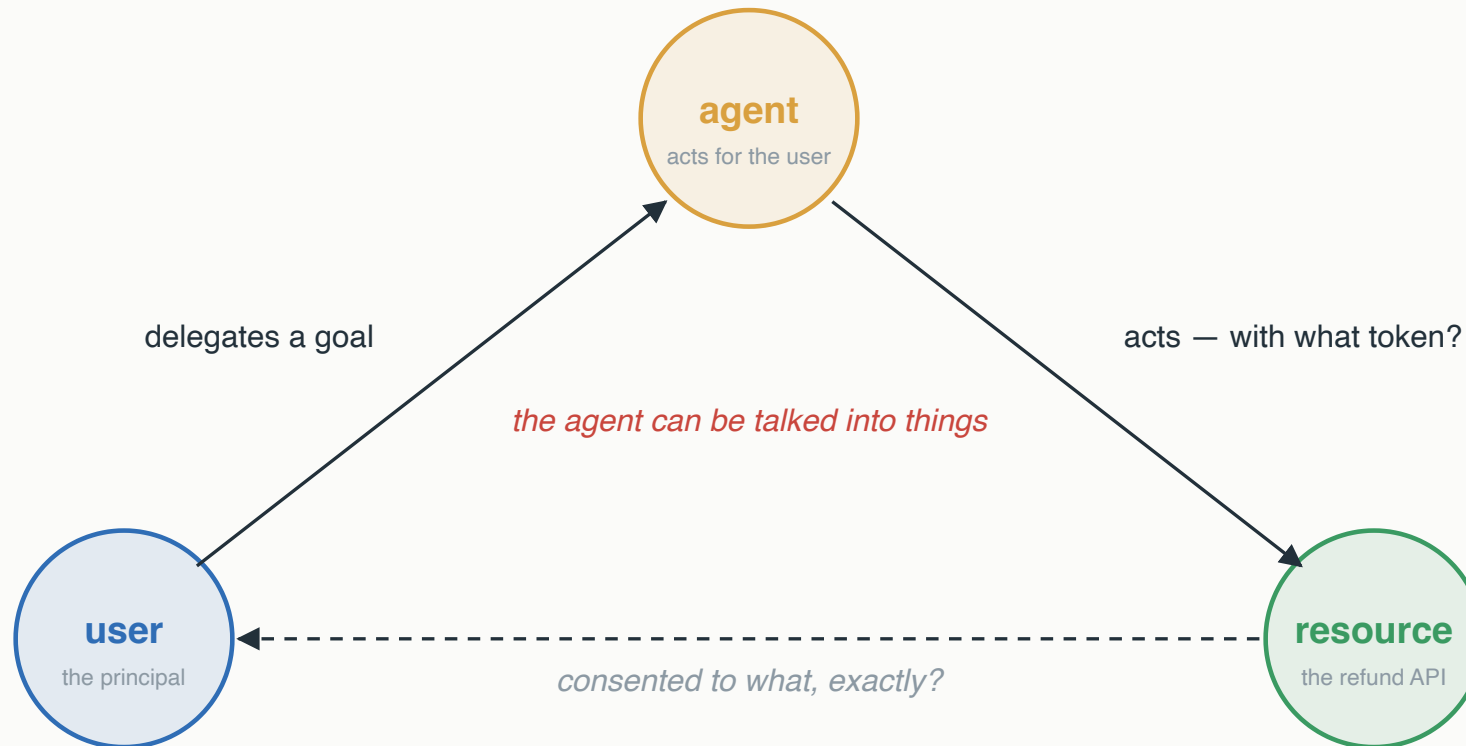
That's the difference between a chatbot and an agent — and the system you were handed tonight *has hands*.

Instructions and data share a channel



Instructions and data arrive through the same channel. The reader cannot reliably tell which lines are orders.

Trust has three parties



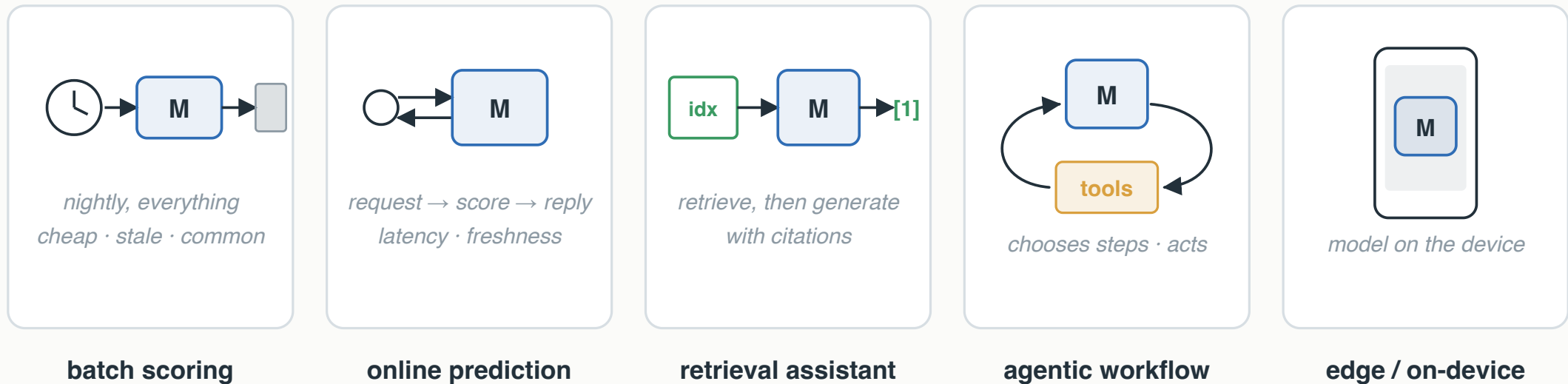
Classic authorization has two parties. An agent has three. Nearly everything hard follows from that.

The Contrast Table, first edition

CLASSIC PRACTICE	AI SYSTEMS PRACTICE
Unit test	Eval harness: sampled suite, slices, statistic, noise floor
Assertion	Runtime guardrail on probabilistic output
CI gate: "tests pass"	CI gate: "did not regress beyond the noise floor"
Version the code	Version the <i>artifact</i> : code + prompts + model + index + tools + eval suite
Types and interfaces	Schemas consumed by a caller that may ignore them
Logging	Tracing a tree of nondeterministic steps
Step-through debugging	Slicing, distribution comparison, reproduction under sampling
Roll back the deploy	Roll back to <i>which</i> combination?

◆ *Engineers: your list of instincts about to betray you. Data scientists: your list of things you hoped someone else would handle.*

Five shapes



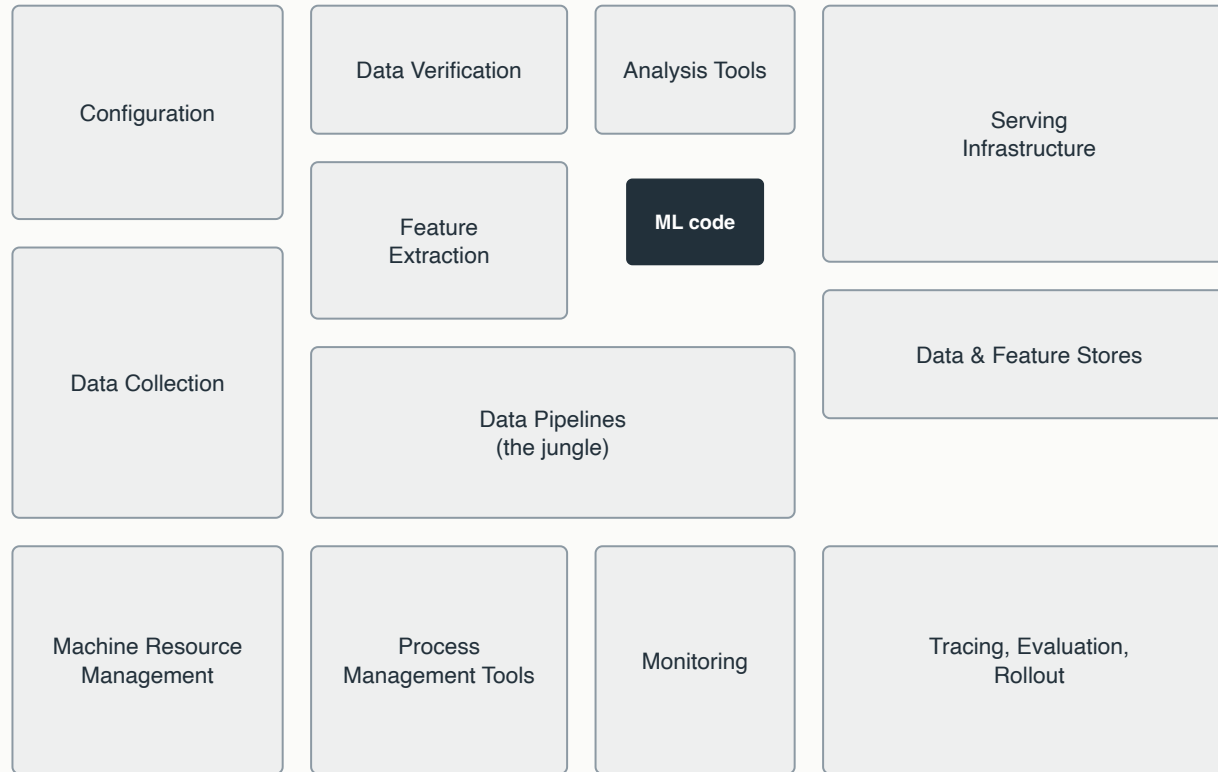
Five shapes. Most systems are one of these, or two of them glued together.

Three running systems

	SHAPE	WHAT IT DOES	WHAT MAKES IT HARD
Scorer	Prediction	Fraud risk on transactions; batch + online	Rare positives, delayed labels, adversaries, a threshold someone owns
Advisor	Grounded generation	Answers over an org's documents, with citations	Retrieval is the bottleneck; freshness; untrusted docs
Operator	Action	The ticket agent you were handed tonight	Untrusted text in context; irreversible refunds; 40-step trajectories; three parties

You build the Operator. The other two are how the same idea lands when the component scores, or writes.

Hidden technical debt



After Sculley et al. (2015), Figure 1.

CACE — Changing Anything Changes Everything. No local changes.

Pipeline jungles. Grown by accretion, owned by nobody. Week 4.

Feedback loops. Third mention. Week 6.

Sculley et al., 2015. A decade old, about scorers. Still right. Agents add property 4 and everything gets worse.

◆ *The tiny box is what everyone wants to work on. The big rectangle is what pages you.*

Which step did Air Canada skip?

Step 1. Nobody wrote down an acceptable rate of wrong policy answers, or who owned the ones that got through.

Step 3. No metric for *groundedness*. It linked to the correct policy while contradicting it. Measurable. Unmeasured.

Step 7. The remainder policy was "argue the chatbot is a separate legal entity." That's not a mitigation. That's a hope.

And what wasn't the problem: the model.

It did what models do. It produced a fluent, plausible sample.

The system around it had no requirements, no measurement, and no plan for being wrong.

The system, not the model.

"Design a system that flags fraudulent transactions."

Scope only — step 1 of the framework, and the first half of step 2: users, the decision, the three numbers, what "good" means and for whom, what happens when it's wrong.

No boxes. No models. No architecture. If you draw a box, I will ask you who the users are.

Build in pairs · then review as a group

Then: the envelope.

Write it up in a document — one page — and **save it somewhere you'll find it again**. Don't polish it.

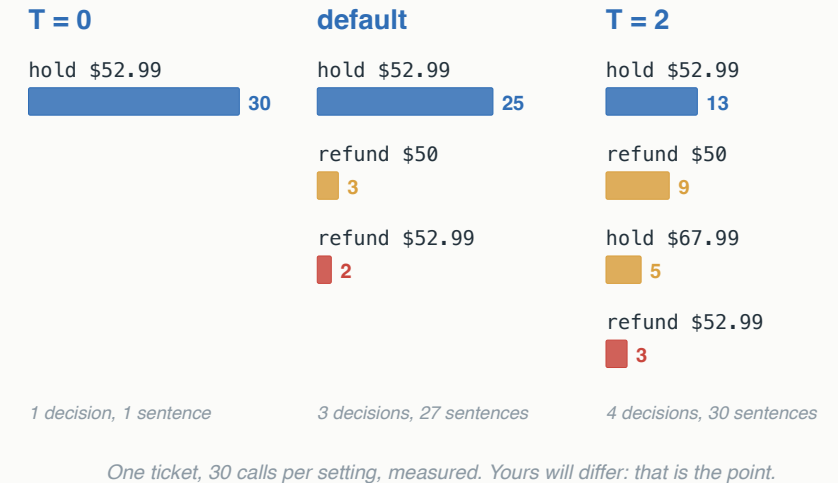
In Week 13 you'll open it and redesign it with everything the course gave you. The gap is the course.

Setup, then feel the distribution

0 — Setup. Clone the lab repo, `uv sync`, Gemini key in `.env`, one each. `distribution.py --n 3` prints three decisions: you are set. **Nothing on AWS tonight.**

1 — The distribution. `distribution.py`: one ticket, 30 calls each at T=0, the default, and T=2. Watch the samples scroll. *How many distinct decisions at T=0? How many distinct sentences? Is the modal answer right? At T=2, which outputs break the policy?*

2 — Find the slice. `slides.py`: find the category that fails $\geq 30\%$, then the ticket inside it. Read the model's reason on the failures. *If setup ate the time, do this before the homework.*



Keep: the plot, your distinct-decision count at T=0, the failing slice. The homework will ask.

Before you go

- **Reading for next week:** Sculley et al., *Hidden Technical Debt in Machine Learning Systems* · Kaestner, *Machine Learning in Production*, Ch. 6 *Gathering Requirements*, Ch. 7 *Planning for Mistakes*, Ch. 27 *Safety* · NIST AI RMF core (skim)
- **P1 is assigned next week**, with the scaffold. Twelve hours, three weeks, pairs.
- Next week: *you cannot specify "correct." You can specify a rate — and then design for the remainder.*
- ◆ *And read the changelog. Somebody has to be the first.*

Homework and Project

- **Homework 1:** Posted in the course guide after tonight's block, due before next week's block.
- **Project 1, *Measure it*, is assigned next week** with the scaffold. Due Week 4.
- **Before next week:** complete the AWS setup checklist in the course guide, under *Supplemental* → *Configuring AWS*. About an hour.